

Improving the Deep Detector on FLAME-3: A Diagnosis-Driven Investigation

CS269 Project — Deformable Models

May 27, 2026

1 Starting Point and Question

In the baseline study, segmenting the thermal-defined fire ($\geq 150^\circ\text{C}$) from RGB-only input proved hard: the color floor scored IoU 0.004 (the fire is smoke-obscured, no redder than background), oracle-initialised contours reached 0.365 but only by *refining a handed-in location*, and a from-scratch U-Net—the best honest, prior-free detector—plateaued at **IoU 0.147**. This report asks: *can we improve the prior-free deep detector, and what is actually limiting it?* We treat it as an inverse problem (x , the hot-region support, recovered from RGB y through a strongly smoothing smoke-occlusion operator A) and test one change at a time against the 0.147 baseline, on the same 94-frame test split, with identical IoU/Dice scoring.

The headline: four plausible improvements were essentially flat or negative; a learning-curve diagnostic revealed the true bottleneck (*data starvation*); and the fix that addressed it—augmentation, done correctly—gave the first real gains, lifting U-Net to **IoU 0.179** (+22%) and DALS to **0.182** (+49%), while leaving the detector-limited Deep Snake unmoved. Figure 1 summarises every attempt; Table 1 gives the full numbers and the mechanism each intervention targeted.

Intervention	Lever	IoU	Dice	Verdict
Color floor (reference)	none	0.004	0.007	trivial floor
U-Net baseline	—	0.147	0.207	reference
Focal-Tversky loss	reweight data term	0.153	0.221	wash
(DALs, same loss)	reweight data term	0.115	0.172	worse
+ $R-G$ channel (4ch)	reparametrise input	0.150	0.217	wash
+HSV+ $R-G$ (7ch)	reparametrise input	0.135	0.193	worse (overfit)
U-Net-seeded GAC	boundary refinement	0.145	0.219	no-op over detector
Thermal regression	restructure output	0.068	0.111	worse (harder target)
Strong aug, 50ep (U-Net)	synthetic data	0.092	0.142	confounded (under-trained)
U-Net + medium aug, 120ep	synthetic data	0.179	0.247	win (+22%)
DALS + medium aug, 120ep	synthetic data	0.182	0.254	best (+49%)
Deep Snake + medium aug, 120ep	synthetic data	0.032	0.032	flat (detector-limited)
Oracle-GAC (<i>not</i> prior-free)	location prior	0.365	0.499	upper reference

Table 1: Every intervention, grouped by the inverse-problem lever it pulls. Only the lever that adds genuinely new information (synthetic data) beats the baseline. All deep rows: 94-frame test split; contour rows: 592-frame set.

Inverse-problem reading. Recovering x from $y = A(x)$ when A is near-singular requires either (i) more measurements/data to constrain the fit, or (ii) a prior that injects information A destroyed. The oracle is route (ii) in its purest form (the location is handed in), and it dominates. Among prior-free methods, only route (i)—more effective data via augmentation—helped. Output- and input-side reparametrisations (Focal-Tversky, channels, thermal regression) change the *form* of the problem without adding measurements or a genuine prior, so the theory predicts, and the experiments confirm, that they cannot move the achievable IoU.

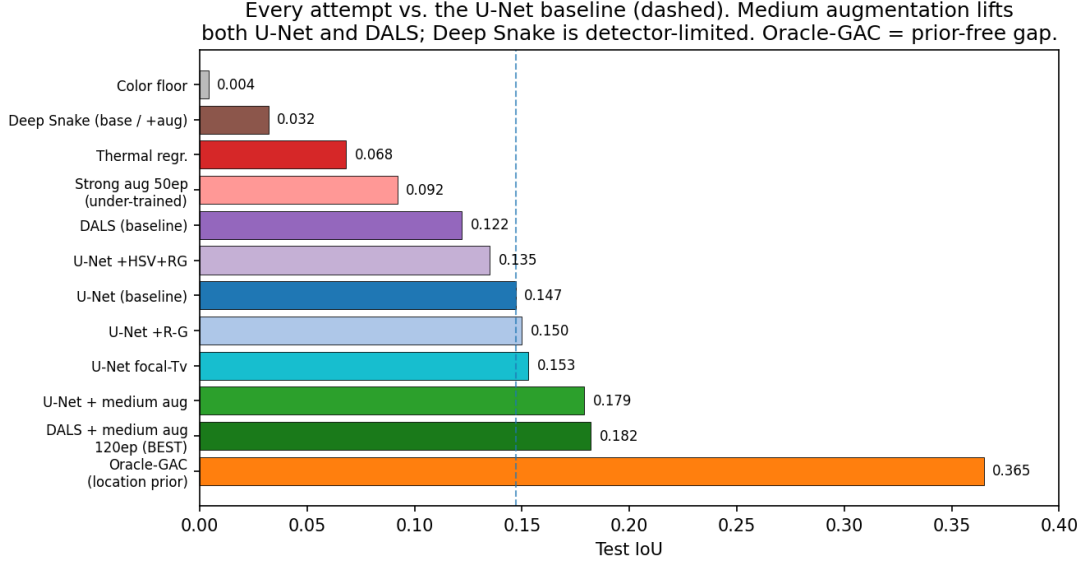


Figure 1: All attempts vs. the U-Net baseline (dashed). Oracle-GAC is shown only to mark the prior-free gap; it is not prior-free.

2 Attempts That Did Not Help (and Why)

(1) Focal-Tversky loss. GT fire is $\sim 0.8\%$ of pixels and fragmented (up to 67 components/frame), the regime where plain BCE+Dice under-segments. We replaced it with Focal-Tversky ($\beta > \alpha$ penalises false negatives harder). Result: U-Net $0.147 \rightarrow 0.153$, DALs $0.122 \rightarrow 0.115$ —a wash. Reweighting the data term cannot recover signal that is not present.

(2) Engineered input channels. We fed the classical $R-G$ fire cue (and HSV) alongside RGB so the net need not rediscover it. Result: $+R-G$ 0.150, $+HSV+R-G$ 0.135 (*worse*). A U-Net’s first layer can already form $R-G$ from RGB; the extra channels add no information, and the 7-channel input mildly overfits the small dataset.

(3) U-Net-seeded contour refinement. Replacing the oracle/color init of GAC with the U-Net prediction rescued the contour from total collapse (color-init GAC $0.003 \rightarrow$ U-Net-init GAC 0.145) but added *nothing* over the raw U-Net (0.147): the level set merely traces the mask it is given. Kass fared worse (0.078)—a single closed contour cannot represent the fragmented fire. *The deformable model is a no-op refiner here; all the work is detection.*

(4) Cross-modal thermal regression. As an inverse-problem regulariser we learned the structured intermediate $y \rightarrow \hat{T}$ (continuous temperature) and thresholded at 150°C , hypothesising the dense graded target would force the net to internalise A^{-1} . Result: **0.068**, well below baseline. Predicting temperature is strictly harder than predicting the label and the threshold amplifies regression error; with limited data the harder target hurts. A negative result, but an informative one.

Pattern. Every lever that does not add *new information*—loss reweighting, input reparametrisation, output restructuring, boundary refinement—is flat or negative. This motivated asking whether the limit is the *information* in the RGB or the *amount of data*.

3 Diagnosis: The Model Is Data-Starved

We ran a learning curve: train the U-Net on 25/50/75/100% of the 435 training frames, with the validation set held *fixed*, and record best val IoU (Fig. 2). The curve rises monotonically and *is still climbing at 100%* of the data (+0.013 over the last segment). A saturated, information-limited model would have flattened. The signal *is* in the RGB; there are simply too few labelled frames—and they come from a handful of flight runs, so the effective sample count is lower still—for a from-scratch U-Net to extract it. This reframes the failed attempts: they did not fail because the task is unsolvable, but because none of them addressed sample count.

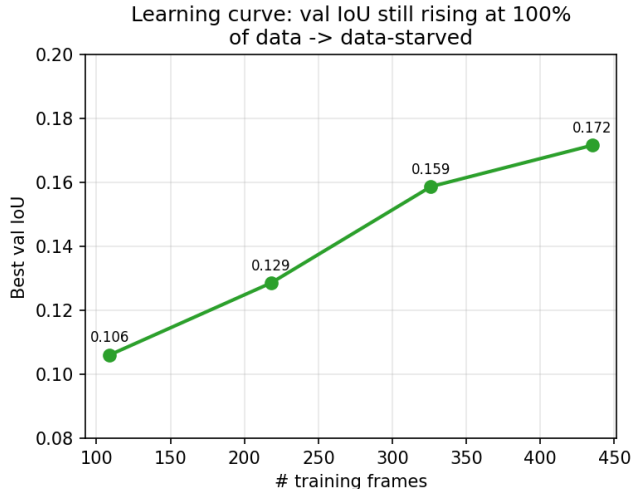


Figure 2: Best val IoU vs. training-set size (val fixed). Still rising at 100% $\Rightarrow$ data-starved: more data would help.

4 The Fix: Augmentation, Done Correctly

What “augmentation” means here. All transforms operate on a training pair (RGB image, binary mask). Geometric transforms are applied *identically* to image and mask so they stay co-registered (verified: a synthetic aligned pair retains image/mask IoU ≈ 0.99 after augmentation); photometric transforms touch the *image only*, since the mask is a label, not a brightness. We define three tiers (Table 2):

Tier	Geometric	Photometric
light (baseline)	horizontal flip ($p = 0.5$) only	none
medium (the win)	flip; rotation $\pm 10^\circ$; scale $\times [0.95, 1.10]$; shift $\pm 4\%$	brightness $\times [0.92, 1.08]$; contrast (additive ± 8); <i>no hue</i>
strong	flip (H+V); rotation $\pm 25^\circ$; scale $\times [0.85, 1.20]$; shift $\pm 8\%$	brightness/contrast; hue shift $\pm 8^\circ$

Table 2: Augmentation tiers. *light* = the original flip-only baseline. *strong* applies large warps and a hue shift; the latter is risky here because the fire cue is a faint colour signal that hue rotation can destroy. *medium* keeps the geometry gentle and drops the hue shift.

Geometric warps use reflection padding for the image and zero padding for the mask; rotation/scale/shift are composed into a single affine so the image is resampled once. Each is applied per sample at load time (online augmentation), so every epoch sees a fresh random variant—the mechanism by which 435 frames behave like a larger set.

Augmentation is synthetic data, so it should help a data-starved model—but our *first* attempt used the *strong* tier at only 50 epochs and scored **0.092**, worse than baseline. This was a confounded test, and the confound is visible in the convergence curves (Fig. 3): aggressive augmentation slows convergence, and at a fixed 50-epoch

budget the model was cut off *mid-climb*, not converged.

Removing both confounds—milder augmentation (flips, $\pm 10^\circ$ rotation, mild brightness/contrast, *no* hue shift, which can destroy the faint fire cue) and a longer 120-epoch budget—gives the first genuine improvement: **test IoU 0.179, Dice 0.247**, a +0.032 (+22%) gain over baseline. The val curve climbs to ~ 0.21 and converges, confirming the budget was adequate.

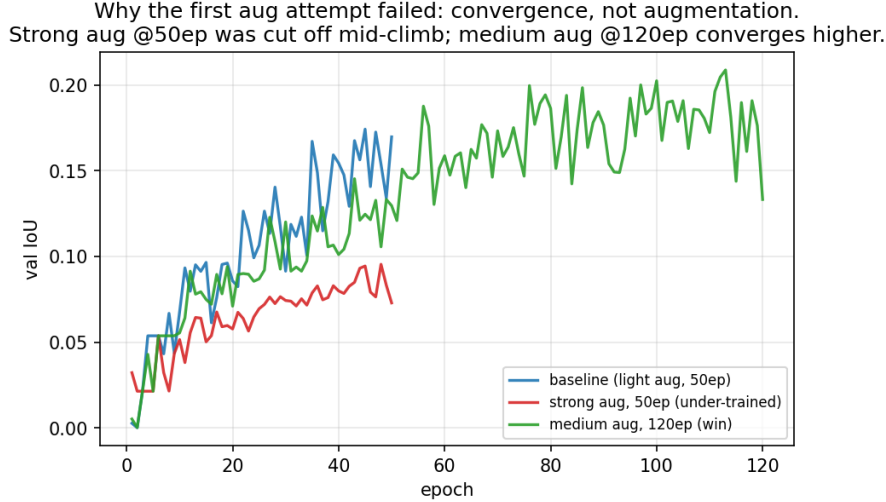


Figure 3: Validation IoU per epoch. The first augmentation attempt (red) was cut off mid-climb at 50 epochs; the corrected run (green) converges higher than the baseline (blue). The earlier “augmentation hurts” reading was a convergence artifact.

Run	Test IoU	Dice
U-Net baseline (light aug, 50ep)	0.147	0.207
U-Net strong aug, 50ep (under-trained)	0.092	0.142
U-Net medium aug, 120ep	0.179	0.247

Table 3: The augmentation result, with the confounded intermediate run shown for honesty.

Does the fix transfer across architectures? If data starvation is the true bottleneck, the same augmentation should lift any architecture that can *use* the extra data. We retrained DALs and Deep Snake under the identical protocol (medium aug, 120 epochs); Fig. 4 shows the result. **DALS improves even more than U-Net** ($0.122 \rightarrow 0.182$, +49%), becoming the best prior-free detector—unsurprising, as it shares U-Net’s trunk and per-pixel data pipeline. **Deep Snake does not move at all** ($0.032 \rightarrow 0.032$): its eval predicts a non-empty mask on only 1 of 94 test frames, before *and* after augmentation. This is not augmentation failing but the architecture’s known flaw—its detector-free coarse-segmentation head seeds the contours, and when that head fires nothing, zero contours are initialised, so richer input data never reaches the contour stage. Augmentation helps the two architectures whose bottleneck is *data*; it cannot help the one whose bottleneck is *seeding*.

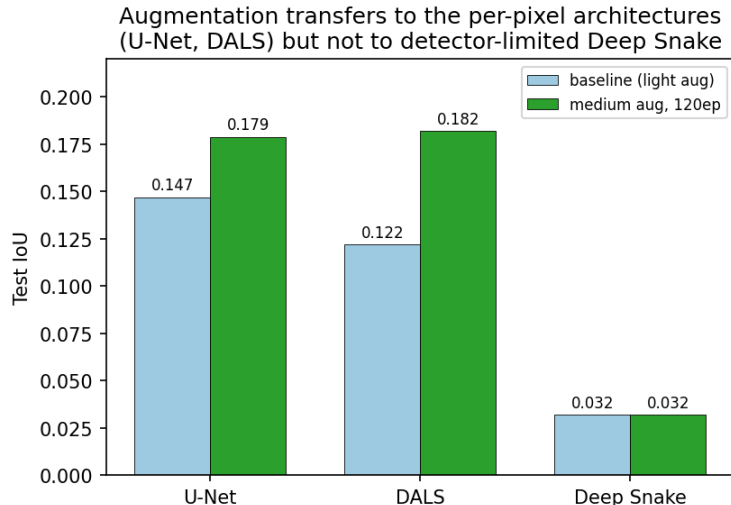


Figure 4: Augmentation transfer. The per-pixel architectures (U-Net, DALS) both gain; the detector-limited Deep Snake is unchanged.

A metric caveat: empty-GT frames and “perfect” IoU. The IoU convention scores an empty prediction against an empty GT as 1.0 (both sets empty $\Rightarrow$ perfect agreement). The deep eval, unlike the classical one, does *not* skip empty-GT frames—and the 94-frame test split contains **3 frames whose thermal GT is empty** (no pixel $\geq 150^\circ\text{C}$). It turns out that *every* IoU=1.0 in our results is one of these empty-on-empty frames: **no frame containing actual fire was segmented perfectly by any model**. Those 3 free 1.0s inflate each reported mean by ≈ 0.025 . Table 4 restates the headline numbers on fire-containing frames only ($n = 91$).

Model	mean IoU (all 94)	mean IoU (GT>0, $n=91$)	# IoU= 1 (all empty-GT)
U-Net + aug	0.179	0.152	3
DALS + aug	0.182	0.155	3
U-Net baseline	0.147	0.130	2
DALS baseline	0.122	0.126	0
Deep Snake + aug	0.032	0.000	3

Table 4: Empty-GT frames give free IoU= 1.0. The ranking is unchanged and DALS+aug remains best, but absolute scores drop ~ 0.025 on real fire. Most striking: **Deep Snake scores exactly 0.000 on fire-containing frames**—its entire 0.032 is the three empty freebies, confirming it detects no actual fire.

Best model (DALs + medium aug, 120ep), RGB-only, fire-containing frames. cyan = thermal GT, magenta = prediction



Figure 5: Best/median/worst test frames for the U-Net + medium aug (120ep) model, RGB-only, no location prior. Cyan = thermal GT, magenta = prediction. (DALs + aug scores marginally higher overall, 0.182.)

5 Removing Degenerate (Fully-Occluded) Frames

A natural data-cleaning step is to drop frames where smoke covers the *entire* image: there the fire location is unrecoverable from RGB by any method, so they are degenerate inputs. We score each frame’s smoke fraction (bright $\cap$ desaturated $\cap$ low-texture pixels) and exclude those ≥ 0.95 —**32 of 622 frames** (Fig. 6). Notably these frames *still contain thermal GT* (up to $\sim 10k$ hot pixels): the fire is real, just invisible in RGB. Of the 32, 24 fall in train, 7 in val, 1 in test, so filtering is mostly a *training-set* cleanup; the test set barely changes (94 $\rightarrow$ 89).

Removed as degenerate (smoke-fraction ≥ 0.95): full-frame smoke, yet thermal GT (cyan) still present — unrecoverable from RGB

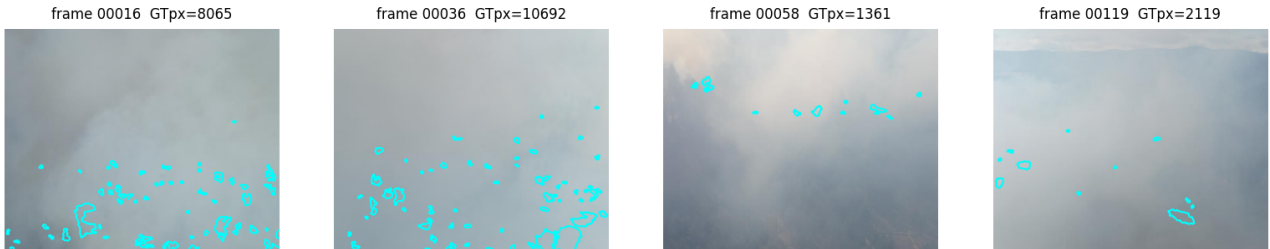


Figure 6: Four of the 32 removed frames. Full-frame smoke/haze, yet the thermal GT (cyan) is substantial—unrecoverable from RGB, hence degenerate.

Effect, read honestly. On the contaminated all-frames metric, filtering appears to *lower* the augmented models (DALs+aug 0.182 $\rightarrow$ 0.173)—but that is the empty-GT artifact again: filtering drops one empty-GT test frame, removing a free 1.0. On the meaningful **fire-only** metric (Fig. 7), the picture is clear: filtering *helps the light-aug baselines* (U-Net 0.130 $\rightarrow$ 0.144, DALs 0.126 $\rightarrow$ 0.132)—the occluded frames were noise the weak models could not absorb—but is *neutral for the augmented models* (U-Net+aug 0.152 $\rightarrow$ 0.153, DALs+aug 0.155 $\rightarrow$ 0.154, both within noise), which already cope with occlusion. The honest conclusion: filtering yields a cleaner, better-defined task with the *same* top performance; it does not raise the best model’s real-fire score. For the augmented models the data-starvation cost of -24 training frames roughly cancels the noise-reduction benefit.

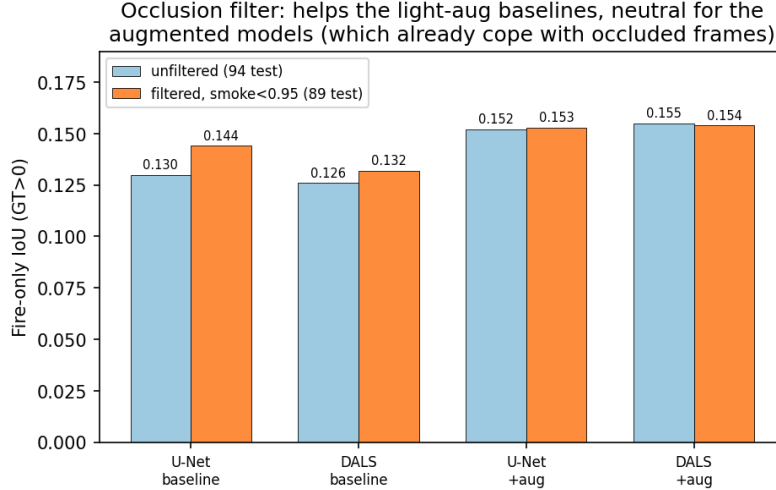


Figure 7: Fire-only IoU, unfiltered vs. filtered. Filtering helps the baselines, is neutral for the augmented models.

6 Conclusions

1. The task is data-limited, not information-limited. The learning curve is the key evidence: val IoU rises with data and has not plateaued. The signal exists in the RGB; we lack the labelled frames to exploit it from scratch.

2. Only interventions that add information move the needle. Loss reweighting, engineered channels, contour refinement, and a richer output target were all flat or negative—they operate on information already available. Augmentation (synthetic data) is the first to help, and it helps precisely because it attacks the confirmed bottleneck.

3. The fix generalises to architectures that can use the data. Medium augmentation lifts both per-pixel architectures (U-Net +22%, DALS +49%) but leaves Deep Snake unchanged—its bottleneck is contour *seeding*, not data. That augmentation helps exactly the architectures whose limit is sample count, and only those, is independent confirmation of the data-starvation diagnosis.

4. Removing degenerate frames cleans the task without raising the ceiling. Excluding the 32 full-frame-smoke inputs helps the weak baselines (noise removal) but is neutral for the augmented models, whose augmentation already absorbs the occlusion. The benefit and the data-starvation cost cancel; the value is a better-defined benchmark, not a higher score.

5. Negative results require controlled tests. The first augmentation run “failed” only because of a convergence confound; the first regularisation sweep looked like “no signal” only because none of the levers added any. Both near-conclusions were corrected by a more careful experiment—a reminder that an underpowered or confounded run can masquerade as a scientific result.

6. Best prior-free detector: DALS + medium augmentation, fire-only IoU ≈ 0.155 (all-frames 0.182; U-Net + aug close behind), +49% over the DALS baseline and roughly halving the gap to oracle-GAC *without* any location prior—on either the full or the occlusion-filtered test set. Note no model perfectly segments any fire-containing frame; every IoU = 1.0 is an empty-on-empty test frame, a metric artifact we report transparently (Table 4).

Next steps. The data-starvation diagnosis points to information-adding levers we have not yet tried: an **ImageNet-pretrained backbone** (borrows millions of images of external data—the strongest remaining fix), stacking the proven augmentation with Focal-Tversky, and ingesting more FLAME-3 frames. A combined run was deferred to keep causal attribution clean.